

4.7 Case Study — Library Management System

لائبریری میں ممبر اور کتاب کا تعلق بہت-سے-بہت سے ہے، اس لیے بورو ریکارڈ نامی پل ٹیبل بناتے ہیں جس میں دونوں کی...

CS XI

Chapter 4: Data and Analysis

Lecture 09 of 15

4.7 Case Study — Library Management System

LEARN IT

A college library needs a database. Walk the seven steps examiners expect.

Step 1 — Requirements. A book has ID, title, author, publisher. A member (student/teacher) has ID, name, contact. A member can borrow many books; a book can be borrowed by different members over time. Record Issue Date and Return Date.

Step 2 — Entities. Book, Member. (Borrowing will need a third entity in step 6.)

Step 3 — Attributes. Book: BookID (PK), Title, Author, Publisher. Member: MemberID (PK), Name, ContactInfo.

Step 4 — Relationship. Member *borrow*s Book.

Step 5 — Cardinality. One member borrows many books over time; one book is borrowed by many members over time → **M:N**.

Step 6 — Resolve M:N. Add junction entity **BorrowRecord**: BorrowID (PK), MemberID (FK), BookID (FK), IssueDate, ReturnDate. Now Member 1—M BorrowRecord M—1 Book (two 1:M links).

Step 7 — Final schema.

```
Member      (MemberID PK, Name, ContactInfo)
Book        (BookID PK, Title, Author, Publisher)
BorrowRecord (BorrowID PK, MemberID FK, BookID FK, IssueDate, ReturnDate)
```



Final library ER: two 1:M relationships through the BorrowRecord junction table.

WORKED EXAMPLE

Sample rows after Sara borrows “Khushwant” on 5 Oct 2026:

BorrowID	MemberID	BookID	IssueDate	ReturnDate
5001	102	B-17	2026-10-05	

The same book can later be borrowed by Ali as BorrowID 5002 — no duplicated book title, no duplicated member address.

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CHECK YOURSELF

1. Why is BorrowRecord needed?
2. Where do IssueDate and ReturnDate belong, and why not on Book?

Answers: 1. To break M:N into two 1:M links. 2. On BorrowRecord — they belong to a *transaction*, not to the book itself (a book is issued many times).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. The library case study starts from:

- a. Requirements
- b. Random tables
- c. Pie charts
- d. Sprints only

MCQ 2. Member to Borrow is typically:

- a. 1:M
- b. Always 1:1
- c. No relation
- d. M:1 the other way only if reversed

MCQ 3. Book to Borrow is typically:

- a. 1:M (one book title/copy can be borrowed many times)
- b. Always M:N without a bridge
- c. 1:1 mandatory
- d. A stack

MCQ 4. M:N Member–Book is resolved by:

- a. Borrow record table
- b. A pie
- c. A NAND
- d. OSI

MCQ 5. A good case-study answer shows:

- a. ER then schema then sample rows
- b. Only careers
- c. Only Python turtle
- d. Only Figma

MCQ 6. Fine calculation would be stored in:

- a. Borrow (or computed in a query)
- b. The OSI header
- c. A K-map
- d. A histogram bin

Extra short questions

1. List four requirements of a college library system.
2. Name the entities.
3. Why a Borrow entity?
4. One report the librarian wants.

Extra long question

1. Walk the seven case-study steps from the notes: requirements → entities → attributes → relationships → keys → resolve M:N → final ER. Draw the final diagram.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Register members; catalogue books; issue/return; calculate fines; search.
2. Member, Book, Borrow (and maybe Copy if multiple copies).
3. The issue event has its own date/fine — and it breaks M:N.
4. Overdue list: member, book, days late.

Long

1. Follow the textbook library walk-through. Final: Member 1–M Borrow M–1 Book. Keys labelled. One sentence on cascade.